From Algorithm to Bit-True Design

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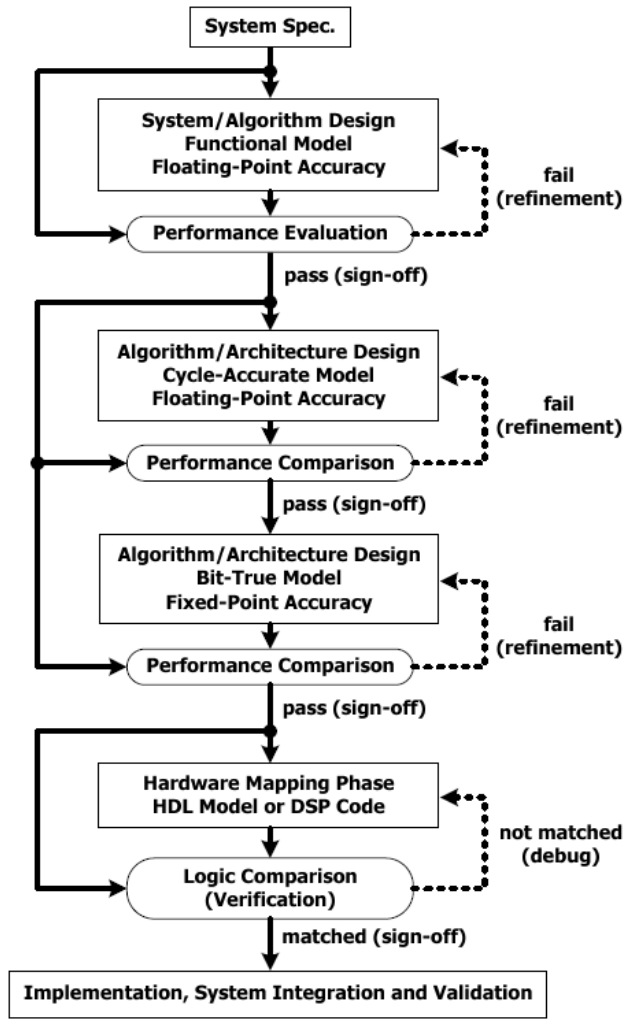
Wireless Communication IC

Digital System Design Flow

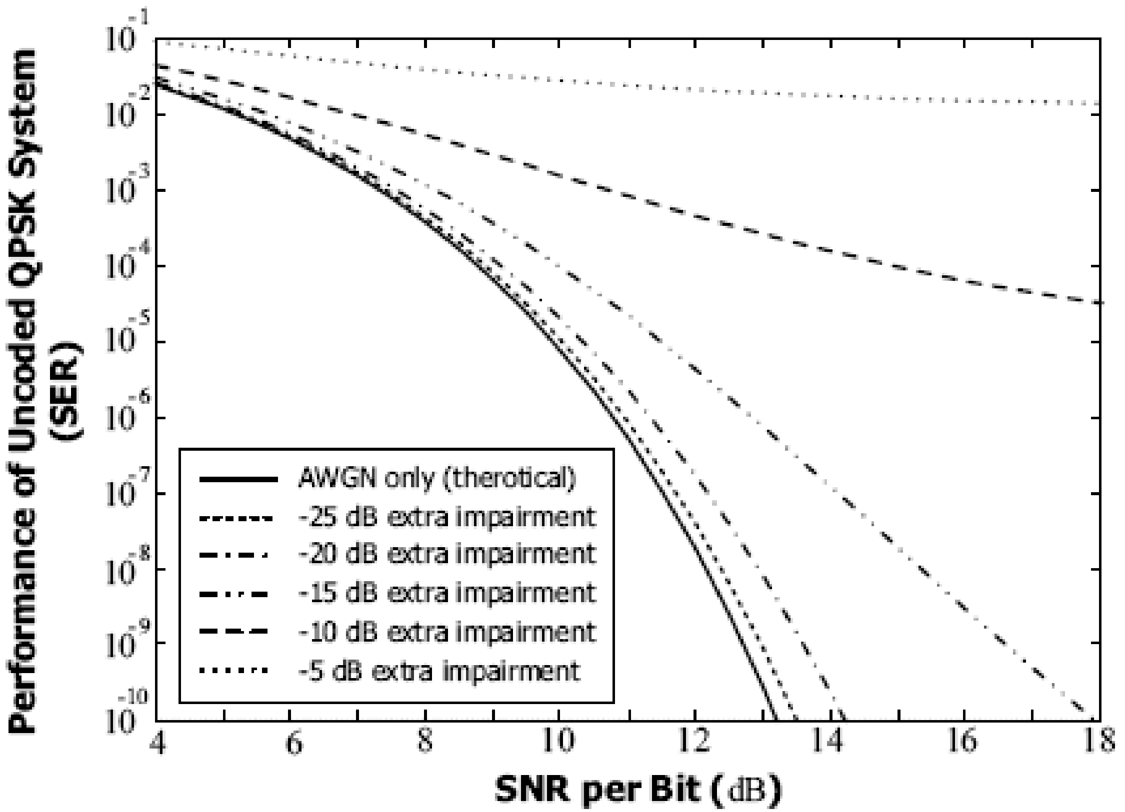


Secondary Impairments in Design

- Source
  - Interference
  - Estimation error in synchronization
  - Finite precision effect (quantization error)
- Often modeled as Gaussian distribution
  - Central limit theorem

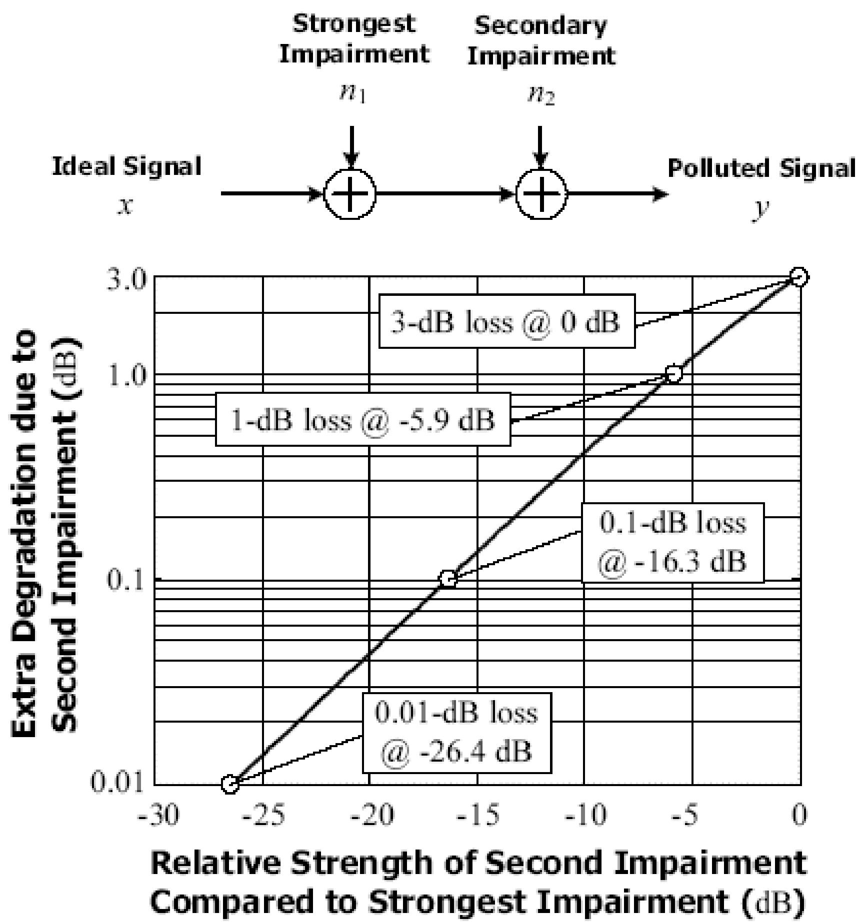
System Performance Degradation

- Introduction of extra impairment with strength relative to signal power



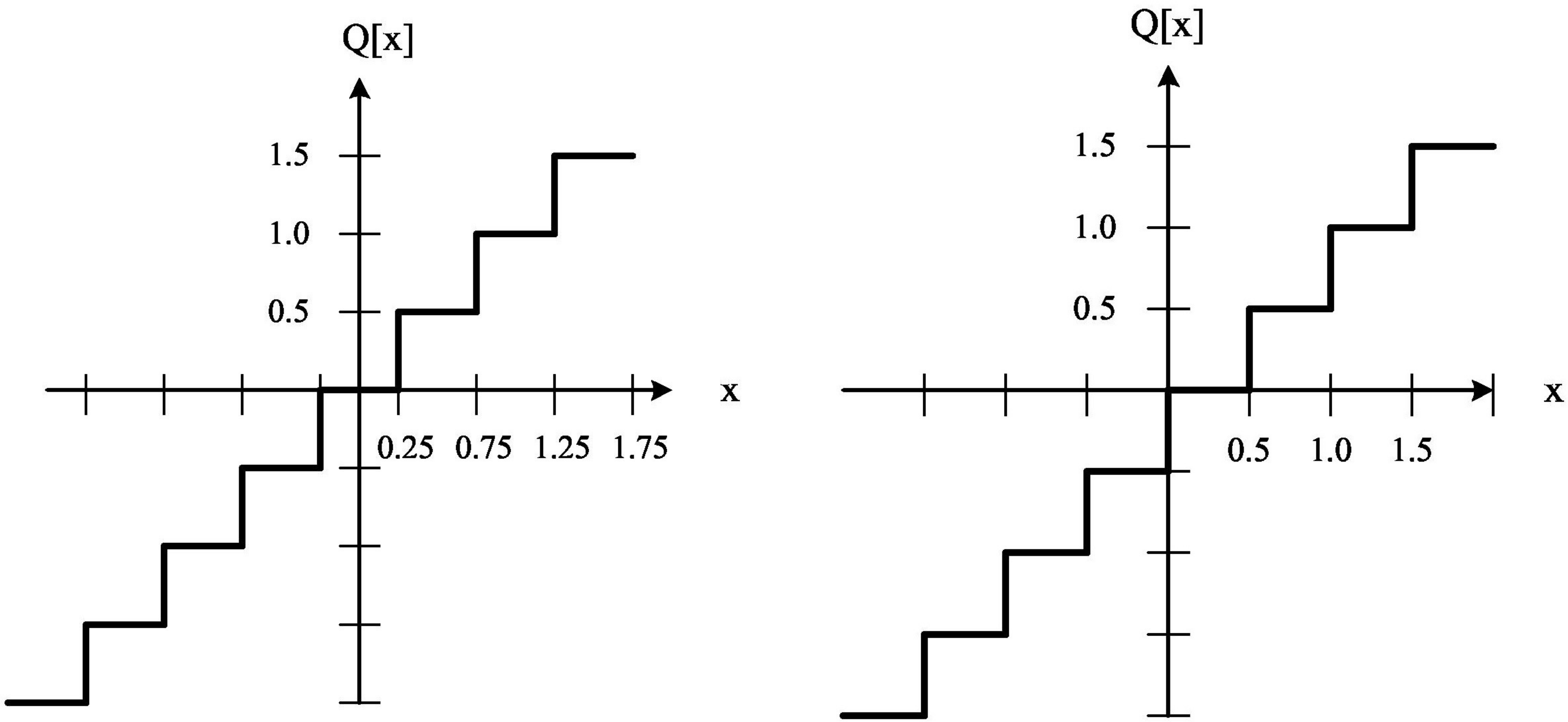
Effects of Additive Impairments

- Extra loss in performance due to secondary impairment

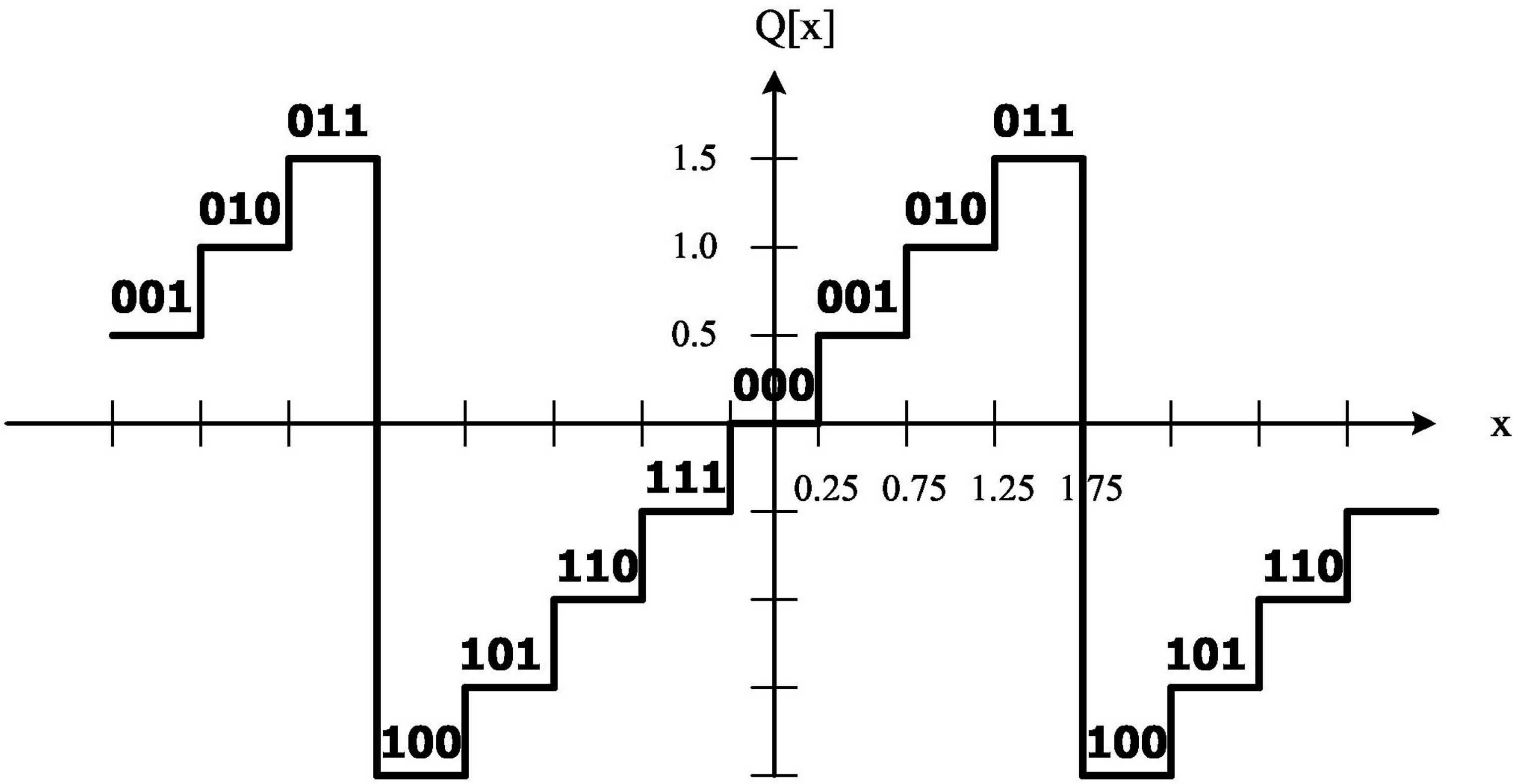


Quantization

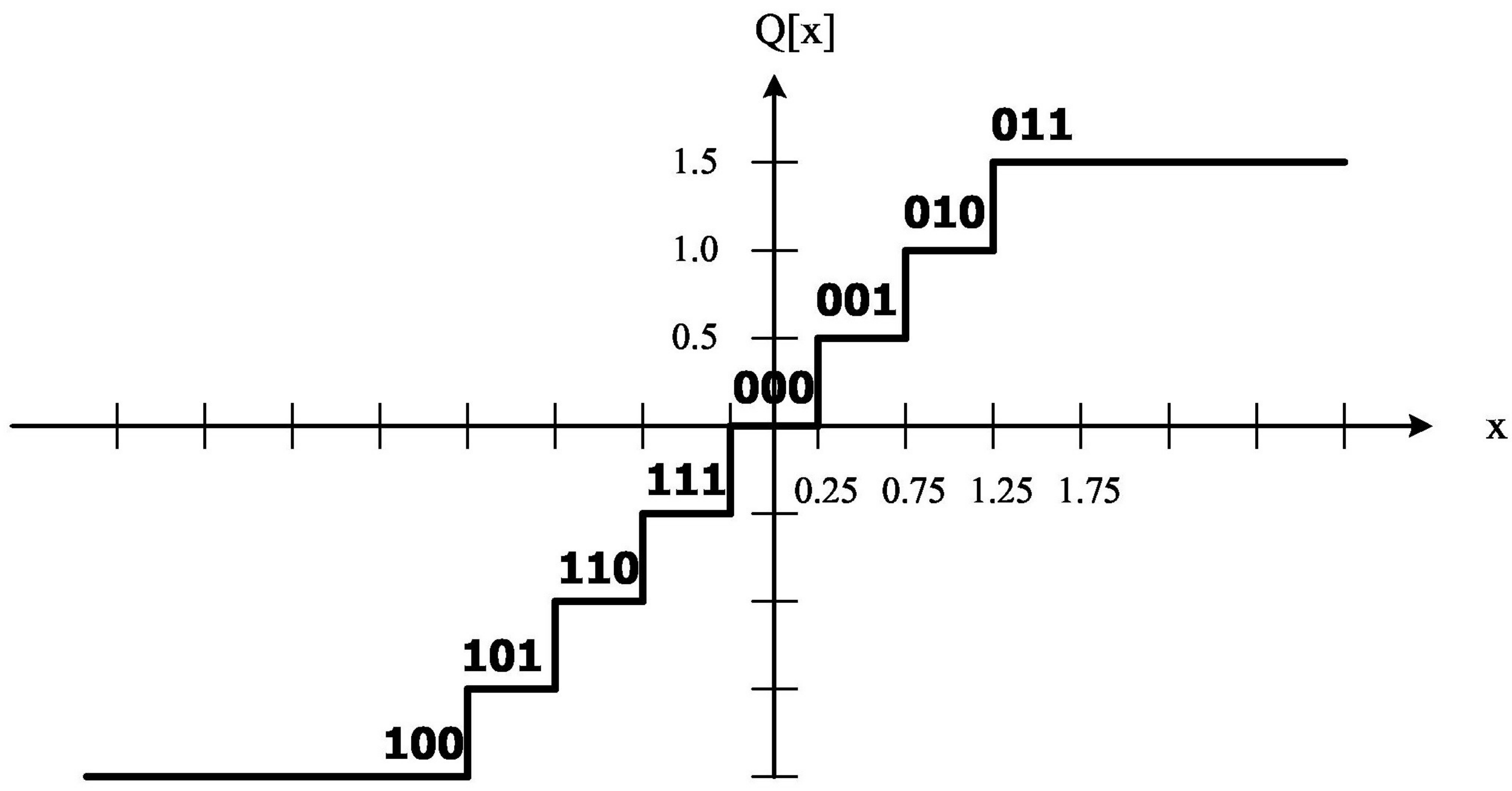
- Rounding v.s. truncation



Overflow (Wrapping)



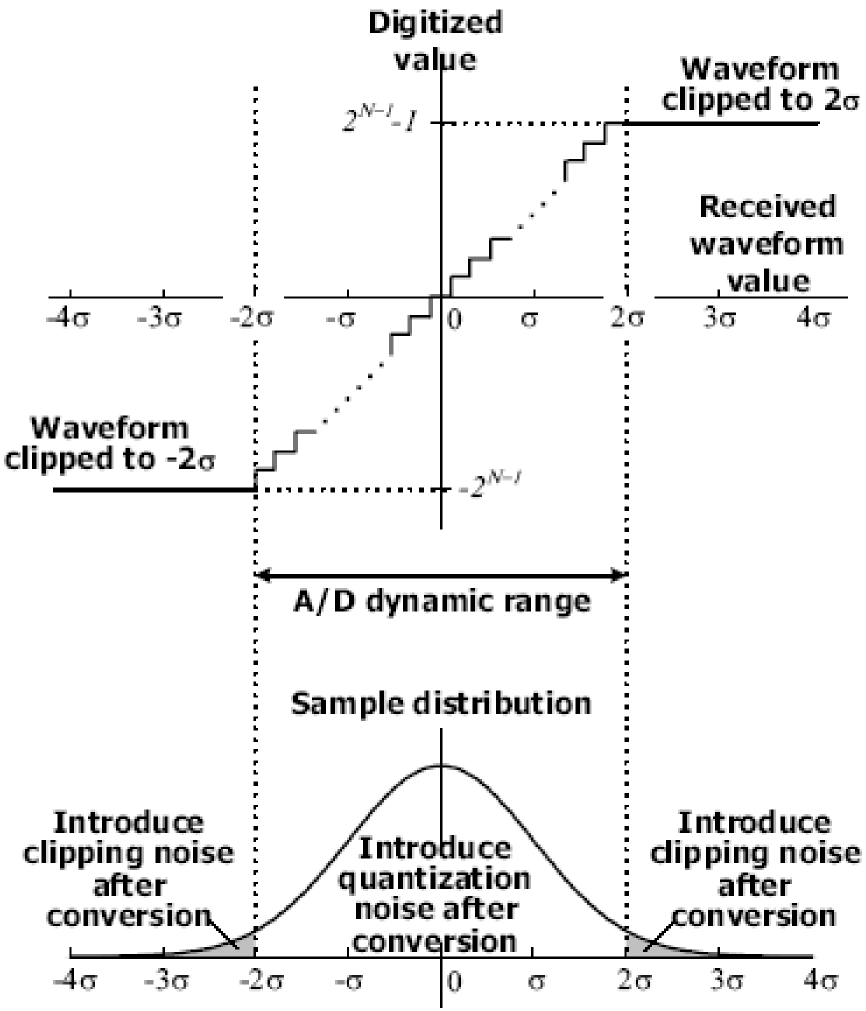
Saturation



I. Analog-to-Digital Conversion

Distortion of ADC (1/2)

- Quantization noise
  - $\Delta = A/2^N$
  - Quantization error uniform distribution in $(-0.5\Delta, 0.5\Delta)$
- Clipping noise



Metrics Related with ADC

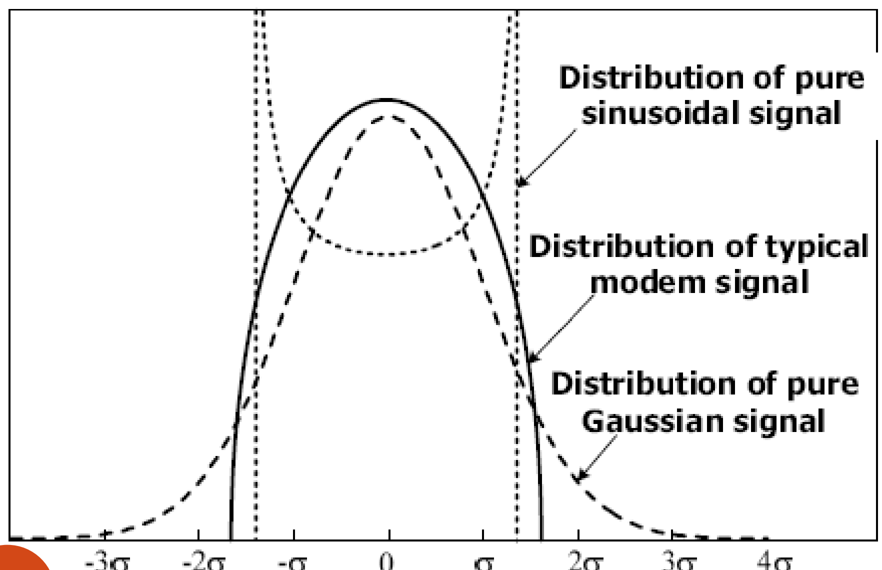
- Effective number of bits (ENOB)
  - Variance of uniform distribution error $\Delta^2/12$
  - Maximum signal power $A^2/8$
  - Signal-to-noise ratio = $10*\log(A^2/(8\Delta^2/12))$

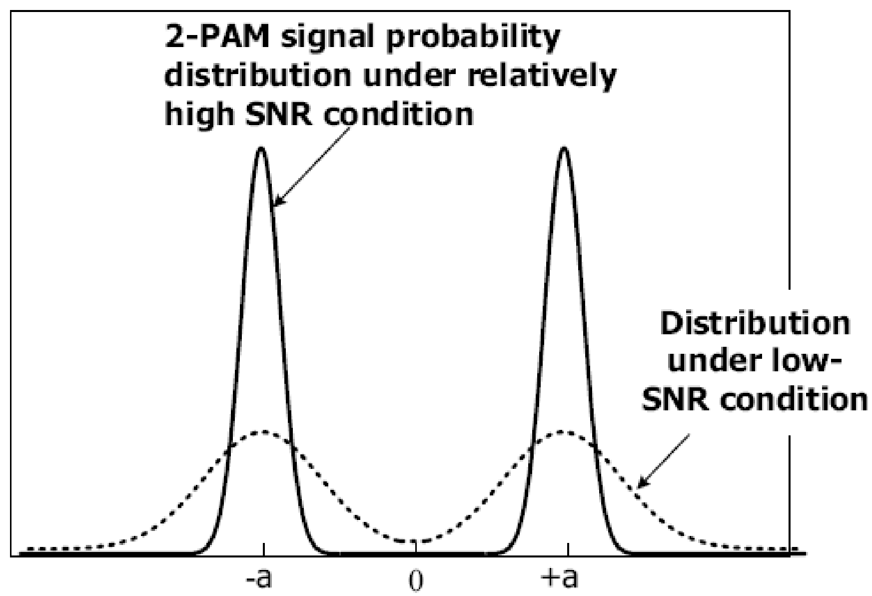
    $= 6.02N+1.76$
  - ENOB = (SNR-1.76)/6.02
- Crest factor (CF)
  - CF=20log(Peak amplitude/RMS amplitude)

Signal Distribution

- Constant amplitude modulation
  - Sinusoidal like
- Typical single carrier signals
- OFDM
  - Gaussian like

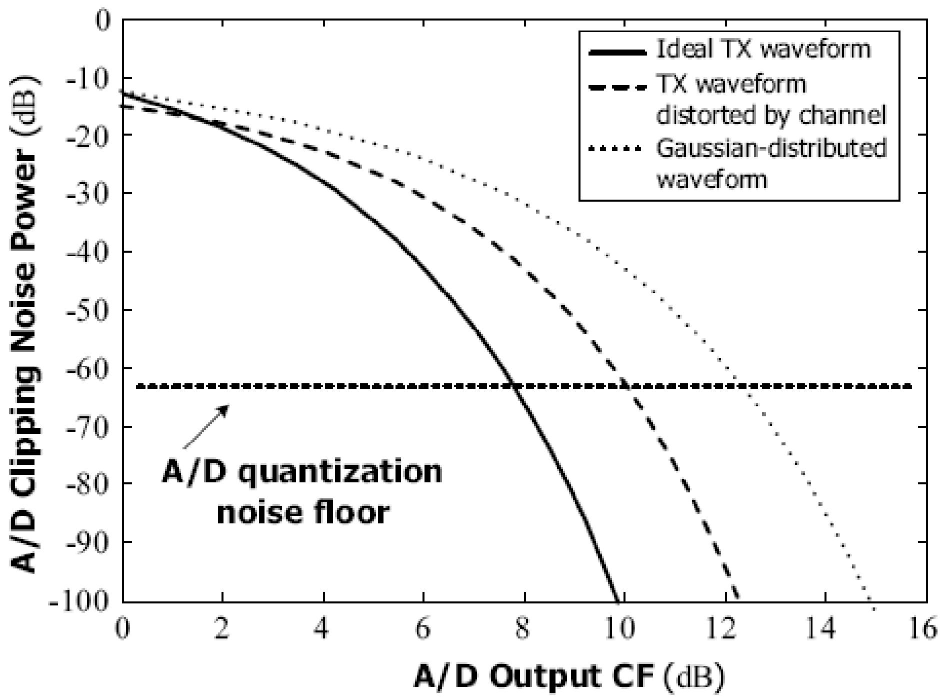
- Multimodal distribution
  - 2-PAM





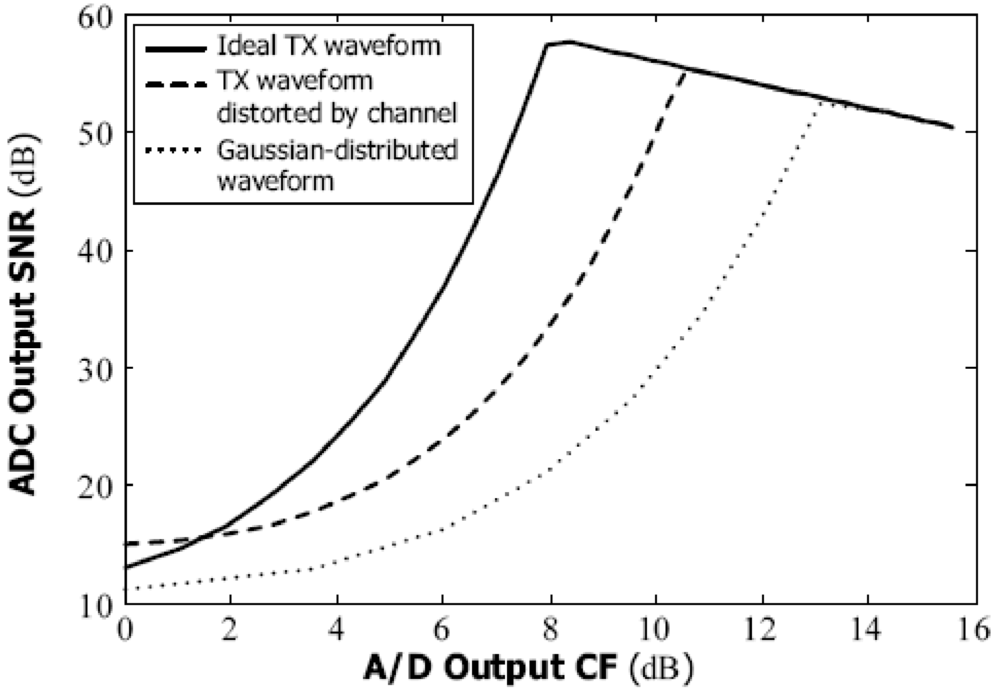
Clipping Noise and CF Setting

- Input is attenuated
  - Clipping noise decreases and output signal CF increases
- More concentrated signal
  - Lower CF at the same clipping noise power



ADC Output SNR and CF Setting

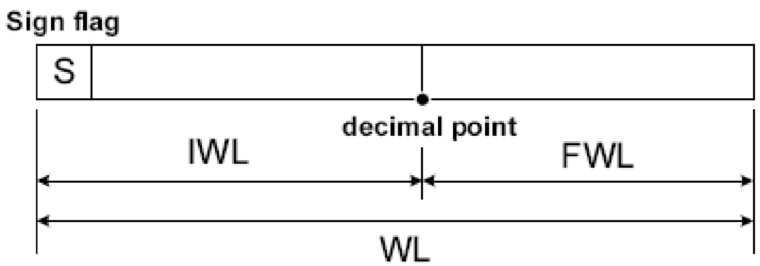
- Attenuation of signal level before ADC can increase output CF
  - Output SNR increase before optimal CF
  - Output SNR decrease after optimal CF



II. Finite Precision Effect

Fixed-Point Data Format

- 3-tuple (WL, IWL, Sign)
  - WL: total wordlength
  - IWL: integer-part
  - Sign: 2's complement or unsigned



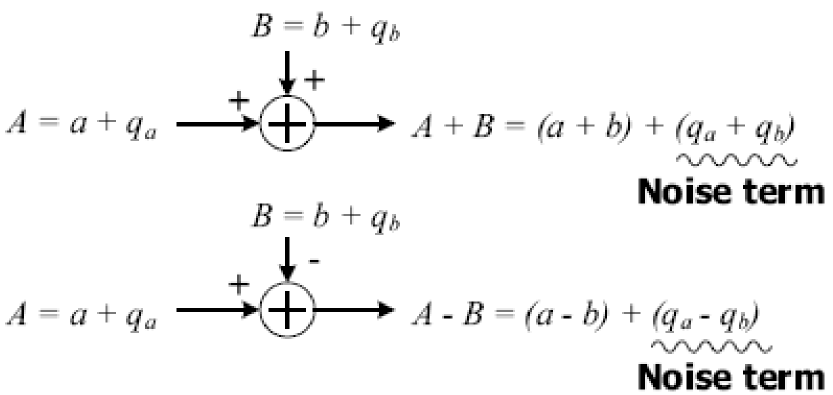
- Alternatives
  - U(MSB,LSB)–--- range [0, 2^{MSB+1}-2^{LSB}]
  - S(MSB, LSB)---- maximum= =(2^{MSB}-2^{LSB}), minimum=- 2^{MSB}

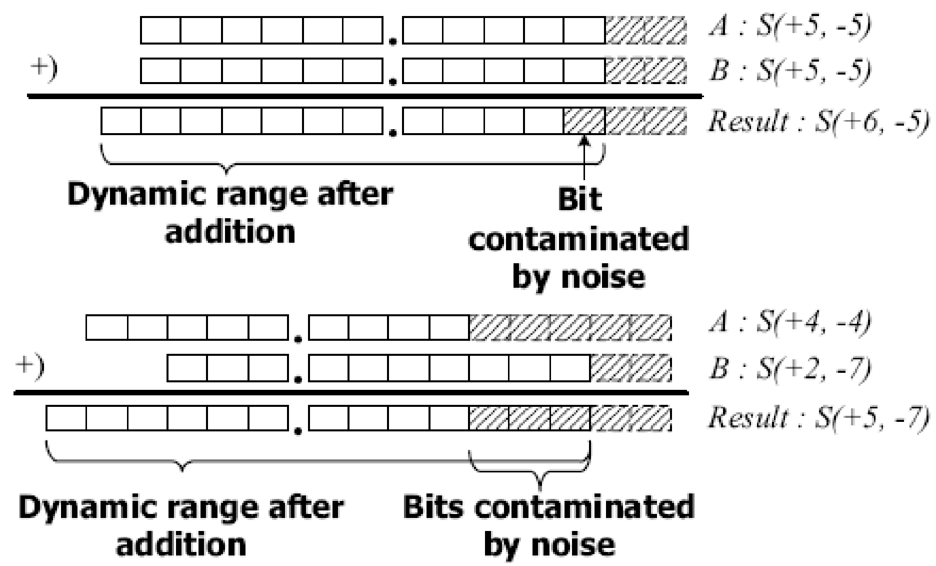
Fixed-Point Error Model

- Truncation error/round error
  - To reduce word-length after multilpication
- Clipping error
  - Usually must be avoided
- Propagation error
  - Error term after arithmetic operation

Propagation Error of Adder/Subtractor

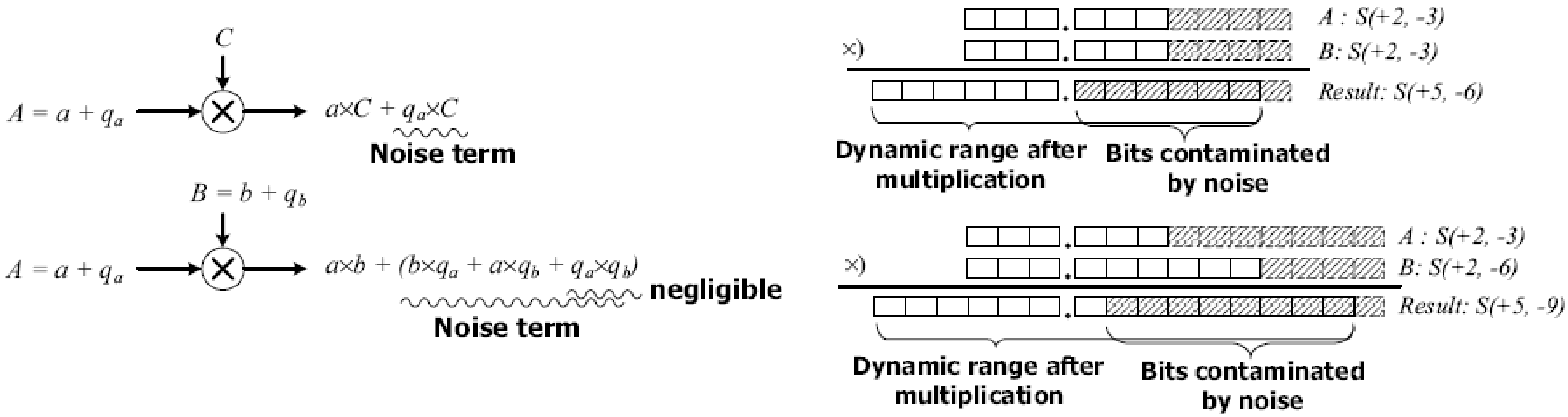
- Error distribution





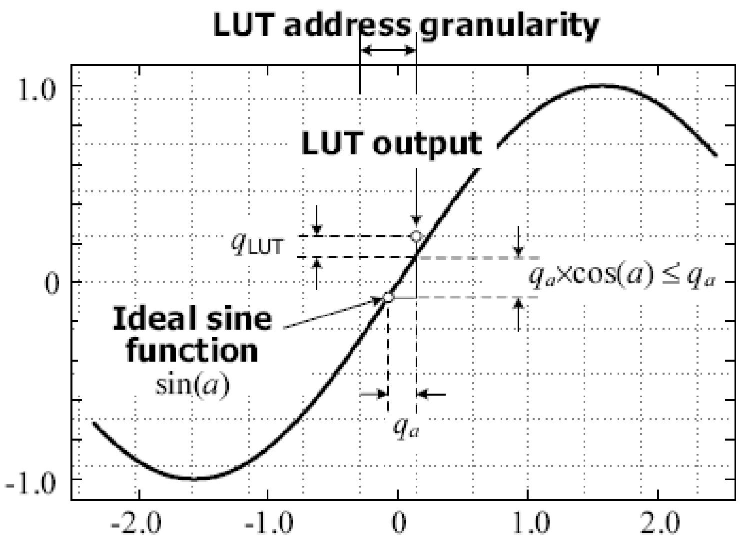
Propagation Error of Multiplier

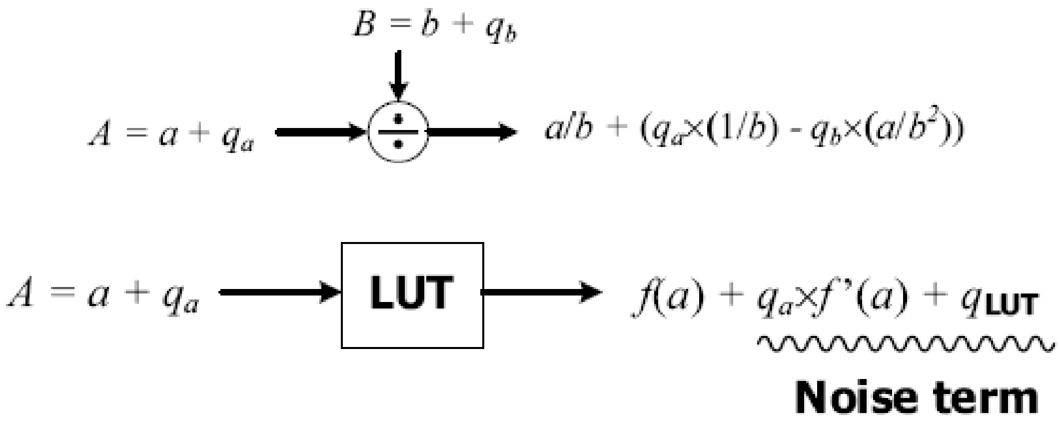
- Error distribution



Other Propagation Error

- Table size
  - Linear increase with output word-length
  - Exponential increase with input word-length





Example

- Sin table with error smaller than 10^{-3}

$$|E_{\text{PROP}}| \approx |q_a f'(a) + q_{\text{LUT}}| \leq |q_a| + |q_{\text{LUT}}|. < 10^{-3}$$

  - Input word-length

$$\text{Input word-length} \geq \log_2\left(2\pi/(2\times 10^{-3})\right) \approx 11.62.$$

  - Output word-length

$$2|q_{\text{LUT}}| < 2\times 10^{-3} - (2\pi/4096) = 4.66\times 10^{-4}.$$

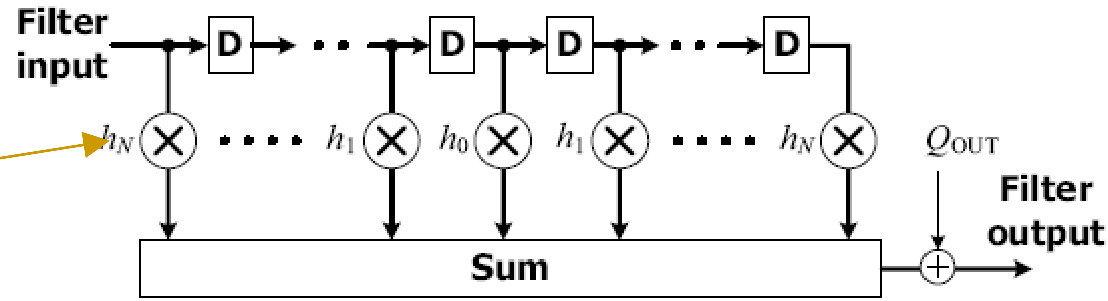
$$\text{Output word-length} \geq 2 + \log_2\left|\frac{1}{2q_{\text{LUT}}}\right| = 13.06.$$

Sign bit + 1 bit IWL

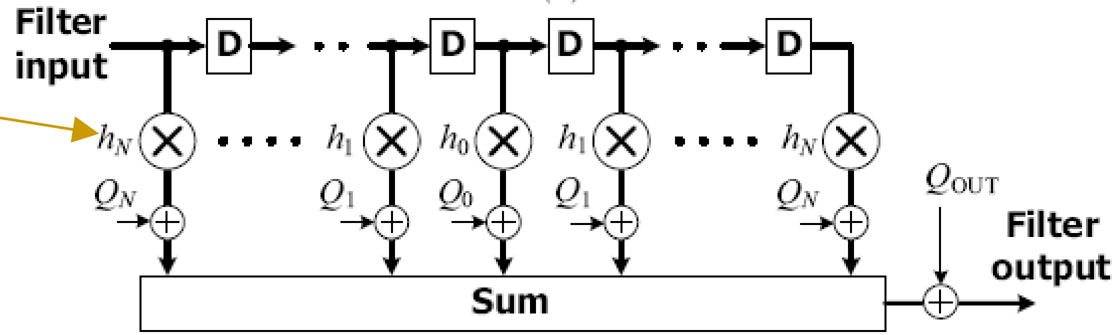
Finite Precision Effects in FIR Filter

- Truncate at filter output v.s. truncate at each product term

Direct form with symmetric coefficients

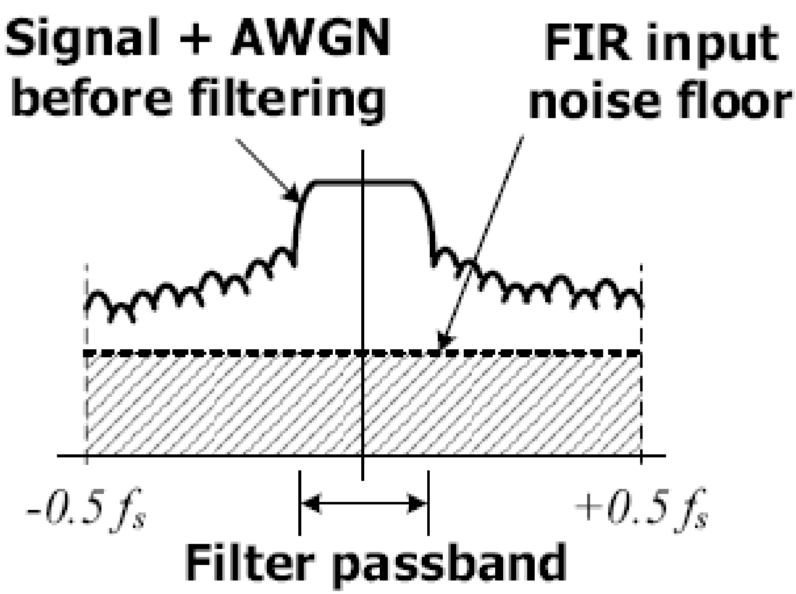


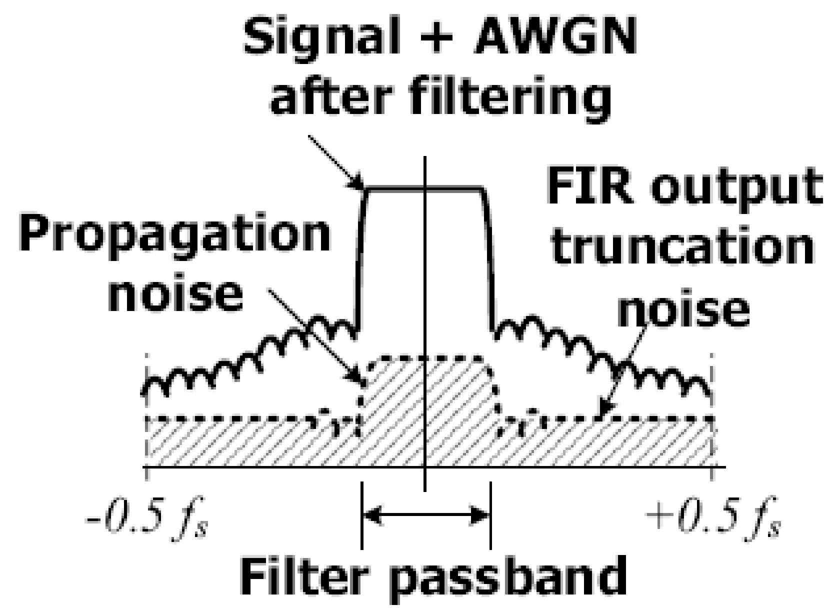
(a)



Filter Output Error

- With input quantization error
  - Propagation error is suppressed in the stop band of the filter.
  - White at input, but not white at output





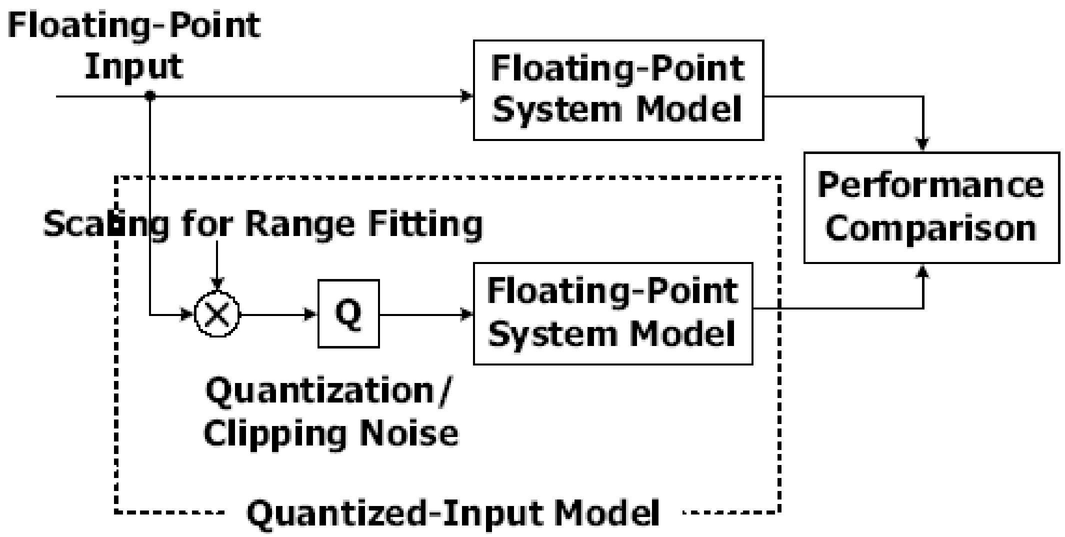
Quantization of FIR Filter Coefficients

- Does not introduce error in data path
- Change filter frequency response
  - Ripple at passband
  - Attenuation at stopband
  - Notch frequency
- Narrow-band band-pass and band-reject filters are sensitive to coefficients

III. From Floating-Point Design to Bit-True Design

Quantized-input Behavior Model

- Scalar (Optional)
- Quantizer
  - Precision
    - Rounding
    - Flooring
  - Clipping
    - Saturation
    - Wrapping

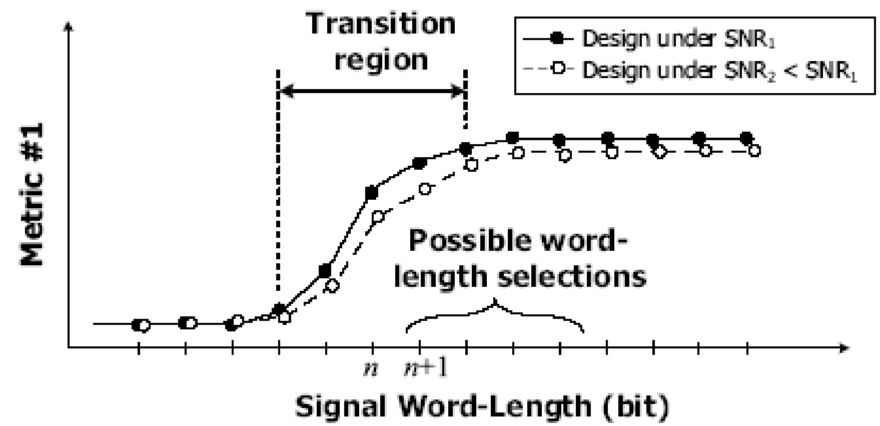


Metric for Performance Evaluation

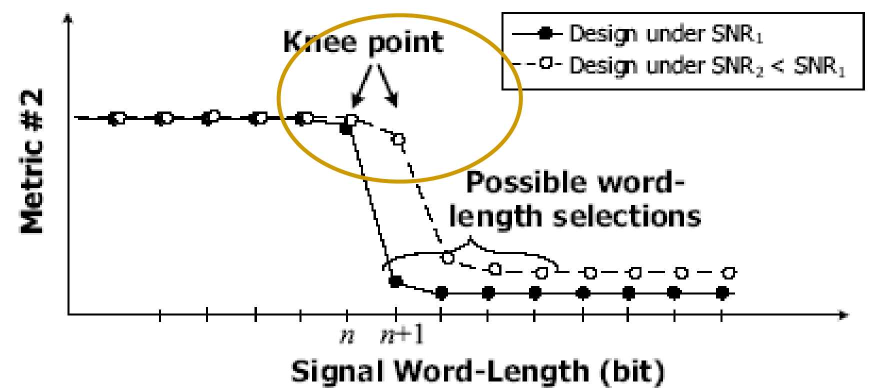
- Waveform difference
  - Suitable for small module
- Signal-to-noise ratio
  - Suitable for the whole system
- Error rate
  - Suitable for evaluation of receiver performance
  - High simulation complexity
- Estimation accuracy
  - Suitable for synchronization/channel estimation

Metric Output v.s. Word-Length

- Output SNR
  - Moderate sensitivity

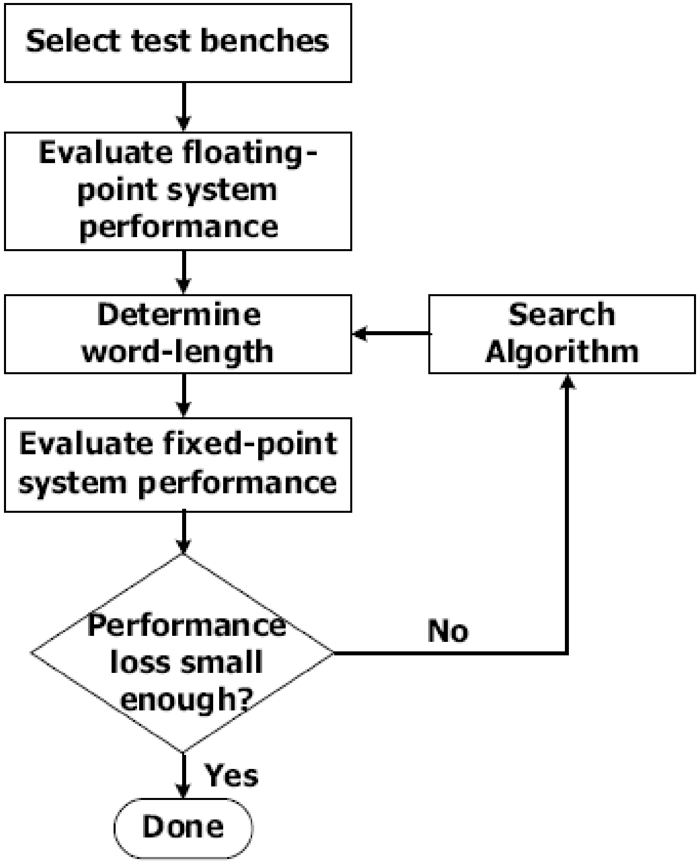


- Error rate
  - High sensitivity
  - Maybe due to error correction codes



Simulation-Based Approach (1/2)

- To be decided in fixed-point representation
  - Integer word-length (by range analysis)
  - Fractional word-length (by precision analysis)
- Simulation for search-based design flow
  - Testbench
  - Cost function
  - Performance metric



Simulation-Based Approach (2/2)

- Cost function
  - Power consumption
  - Area/complexity
- Performance metric
  - Acceptable degradation
- Full search
  - Usually infeasible
- Sequential search
  - Frequently used